

UNIT 1 – ELECTRIC CHARGE, FORCE, AND FIELD

Unit 1 Notes**Answer key**

Placeholder worksheet for the site prototype. Free for classroom use.

Key ideas

- Charge is quantised ($q = ne$, $e = 1.60 \times 10^{-19}$ C) and conserved.
- **Coulomb's law:** $\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r}$, with $k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9$ N·m²/C².
- **Superposition:** the net force (or field) is the vector sum of the individual contributions.
- **Electric field:** $\vec{E} = \vec{F}/q_0$; for a point charge $E = k \frac{|q|}{r^2}$, pointing away from positive charge and toward negative charge.
- A charge in a field feels $\vec{F} = q\vec{E}$; in a *uniform* field the motion is projectile-like with $a = qE/m$.
- **Dipole** in a uniform field: $\tau = pE \sin \theta$, $U = -\vec{p} \cdot \vec{E}$.

Constants

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2) \quad m_e = 9.11 \times 10^{-31} \text{ kg} \quad m_p = 1.67 \times 10^{-27} \text{ kg}$$